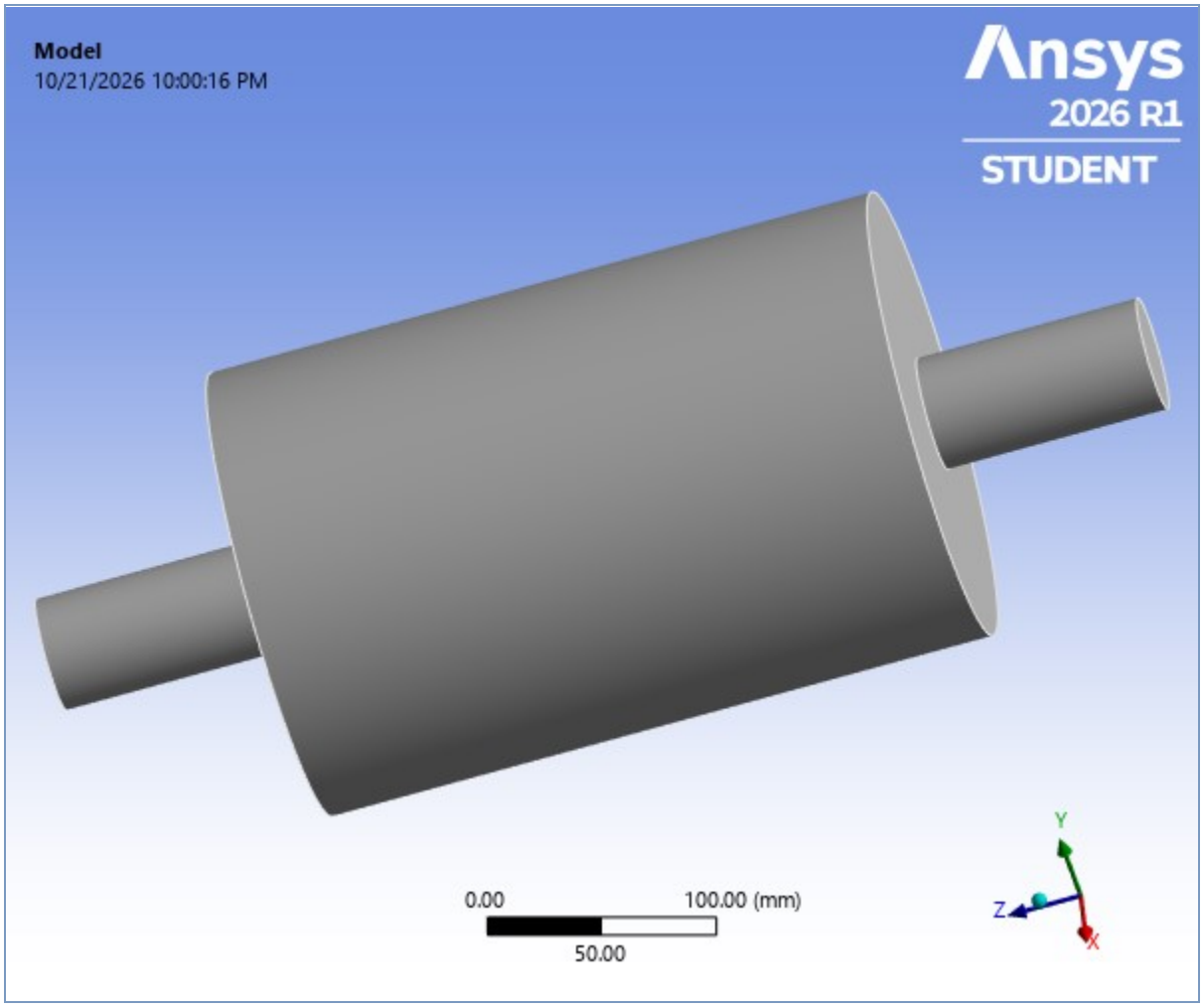
Project*

First Saved	Wednesday, October 21, 2026
Last Saved	Wednesday, October 21, 2026
Product Version	2026 R1
Save Project Before Solution	No
Save Project After Solution	No



Contents

- **Units**
- **Model (A4)**
  - Geometry Imports
    - Geometry Import (A3)
  - Geometry
    - SYS\Solid
  - Materials
  - Coordinate Systems
  - Mesh Workflows
  - Mesh
  - **Harmonic Acoustics (A5)**
    - Pre-Stress/Modal (None)
    - Analysis Settings
    - Acoustics Region
    - Loads
    - Solution (A6)
      - Solution Information
      - Results
- **Material Data**
  - Air

Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2
Model (A4) > Geometry Imports

Object Name	*Geometry Imports*
State	Solved

TABLE 3
Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	*Geometry Import (A3)*
State	Solved
Definition	
Source	D:\Ansys_Temp\WB_User_4544_2\wbnew_files\dp0 \SYS\DM\SYS.scdocx
Type	SpaceClaim
Basic Geometry Options	
Solid Bodies	Yes

Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	D:\Ansys_Temp\WB_User_4544_2\wbnew_files\dp0 \SYS\DM\SYS.scdocx
Type	SpaceClaim
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	200. mm
Length Y	200. mm
Length Z	500. mm
Properties	
Volume	9.8175e+006 mm³
Mass	1.2026e-002 kg
Scale Factor Value	1.
Statistics	
Bodies	1
Active Bodies	1
Nodes	88998
Elements	51335
Mesh Metric	None

Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	*SYS\Solid*
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Air
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	200. mm
Length Y	200. mm
Length Z	500. mm
Properties	

Volume	9.8175e+006 mm³
Mass	1.2026e-002 kg
Centroid X	-3.3796e-007 mm
Centroid Y	2.7795e-015 mm
Centroid Z	249.99 mm
Moment of Inertia Ip1	134.97 kg·mm²
Moment of Inertia Ip2	134.97 kg·mm²
Moment of Inertia Ip3	57.766 kg·mm²
Statistics	
Nodes	88998
Elements	51335
Mesh Metric	None
CAD Attributes	
PartTolerance:	0.00000001
SCRootPartComponent	
Color:143.175.143	

TABLE 6
Model (A4) > Materials

Object Name	*Materials*
State	Fully Defined
Statistics	
Materials	5
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm
Origin Z	0. mm
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

TABLE 8
Model (A4) > Mesh Workflows

Object Name	*Mesh Workflows*
State	Solved
Definition	
Active Mesh Workflow	

Mesh

TABLE 9
Model (A4) > Mesh

Object Name	*Mesh*
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	5.0 mm
Sizing	
Mesh Sizing	Adaptive
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	574.46 mm
Average Surface Area	40374 mm²
Minimum Edge Length	157.08 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (0.1)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	88998

Elements	51335
Show Detailed Statistics	No

Harmonic Acoustics (A5)

TABLE 10
Model (A4) > Analysis

Object Name	*Harmonic Acoustics (A5)*
State	Solved
Physics Definition	
Structural	No
Acoustics	Yes
Definition	
Analysis Type	Harmonic Response
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 11
Model (A4) > Harmonic Acoustics (A5) > Initial Condition

Object Name	*Pre-Stress/Modal (None)*
State	Fully Defined
Definition	
Modal Environment	None
Pre-Stress Environment	None Available

TABLE 12
Model (A4) > Harmonic Acoustics (A5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Step Controls	
Multiple Steps	No
Options	
Frequency Spacing	Linear
Range Minimum	0. Hz
Range Maximum	1000. Hz
Solution Intervals	100
User Defined Frequencies	Off
Solution Method	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Scattering Controls	
Scattered Field Formulation	Program Controlled
Advanced	
Far-field Radiation Surface	Program Controlled
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Volume and Energy	Yes

Euler Angles	Yes
General Miscellaneous	Yes
-- Value	Program Controlled
Calculate Velocity	No
Calculate Energy	No
Result File Options	
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	D:\Ansys1\Mifler VS No Mufler Harmonic Acoustic\Mufler\Complete_File\Final_Analysis_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Solver Units	Active System
Solver Unit System	nmm

TABLE 13
Model (A4) > Harmonic Acoustics (A5) > Acoustics Region

Object Name	*Acoustics Region*
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Structural	No
Acoustics	Yes
Suppressed	No
Acoustic Domain Definition	
Artificially Matched Layers	Off
Advanced Settings	
Reference Pressure	2.e-011 MPa
Reference Static Pressure	0.10132 MPa
Fluid Behavior	Compressible

TABLE 14
Model (A4) > Harmonic Acoustics (A5) > Loads

Object Name	*Port*	*Port 2*	*Port In Duct*	*Port In Duct 2*	*Far-field Radiation Surface*
State	Fully Defined				
Port Surface(s)					
Scoping Method	Geometry Selection				
Geometry	1 Face				
Port Position	On Exterior Face				
Inside Surface Bodies					
Scoping Method	Geometry Selection				Geometry Selection
Geometry	1 Body				1 Body
Definition					

Type	Port	Port In Duct		Far-field Radiation Surface
Port Behavior	Transparent			
Suppressed	No			
Port Attribution		Inlet	Outlet	
Wave Type		Planar Wave		
Coordinate System		Global Coordinate System		
Pressure Amplitude		5.e-005 MPa		
Phase Angle		0. °		
Angle Phi (From X Axis Toward Y Axis)		0. °		
Angle Theta (From Z Axis Toward X-Y Plane)		0. °		
Port Selection		Port	Port 2	
Far-field Radiation Surface Selection				
Scoping Method				Geometry Selection
Geometry				1 Face

Solution (A6)

TABLE 15
Model (A4) > Harmonic Acoustics (A5) > Solution

Object Name	*Solution (A6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	10 m 52 s
MAPDL Memory Used	692. MB
MAPDL Result File Size	1.562 GB
Post Processing	
Beam Section Results	No

FIGURE 1
Model (A4) > Harmonic Acoustics (A5) > Solution (A6)

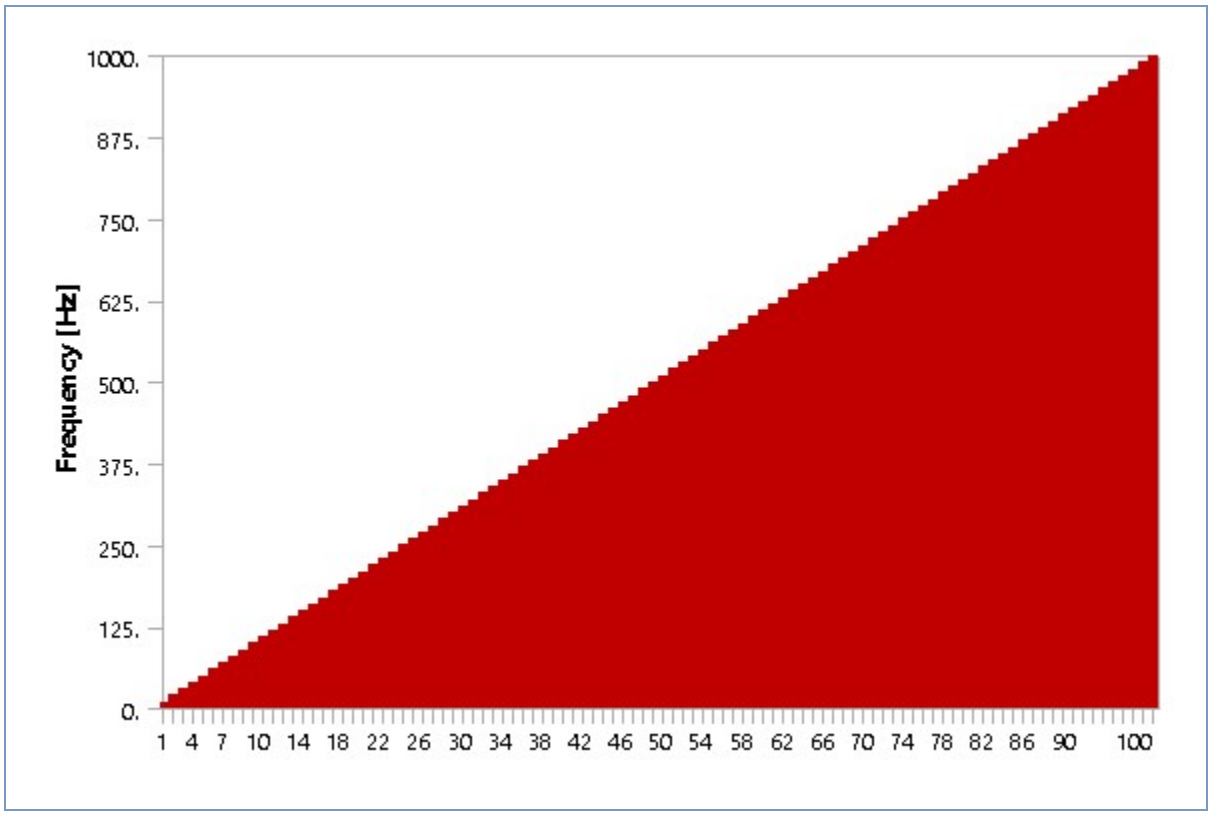


TABLE 16
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 17
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Results

Object Name	*Transmission Loss*	*Far-field SPL Mic*	*Far-field A-Weighted SPL Mic*
State	Solved		
Definition			
Type	Transmission Loss	Far-field SPL Mic	Far-field A-Weighted SPL Mic
Input Port	Port		
Output Port	Port 2		
Suppressed	No		
Definition Method		Coordinates	
Coordinate System		Global Coordinate System	

X Coordinate		0. mm	
Y Coordinate		0. mm	
Z Coordinate		-100. mm	
Reference RMS Sound Pressure		2.e-011 MPa	
Options			
Frequency Range	Use Parent		
Minimum Frequency	0. Hz		
Maximum Frequency	1000. Hz		
Results			
Minimum	0. dB	39.168 dB	-13.483 dBA
Maximum	18.52 dB	57.687 dB	55.329 dBA
Advanced			
Model Type		3D	

FIGURE 2
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Transmission Loss

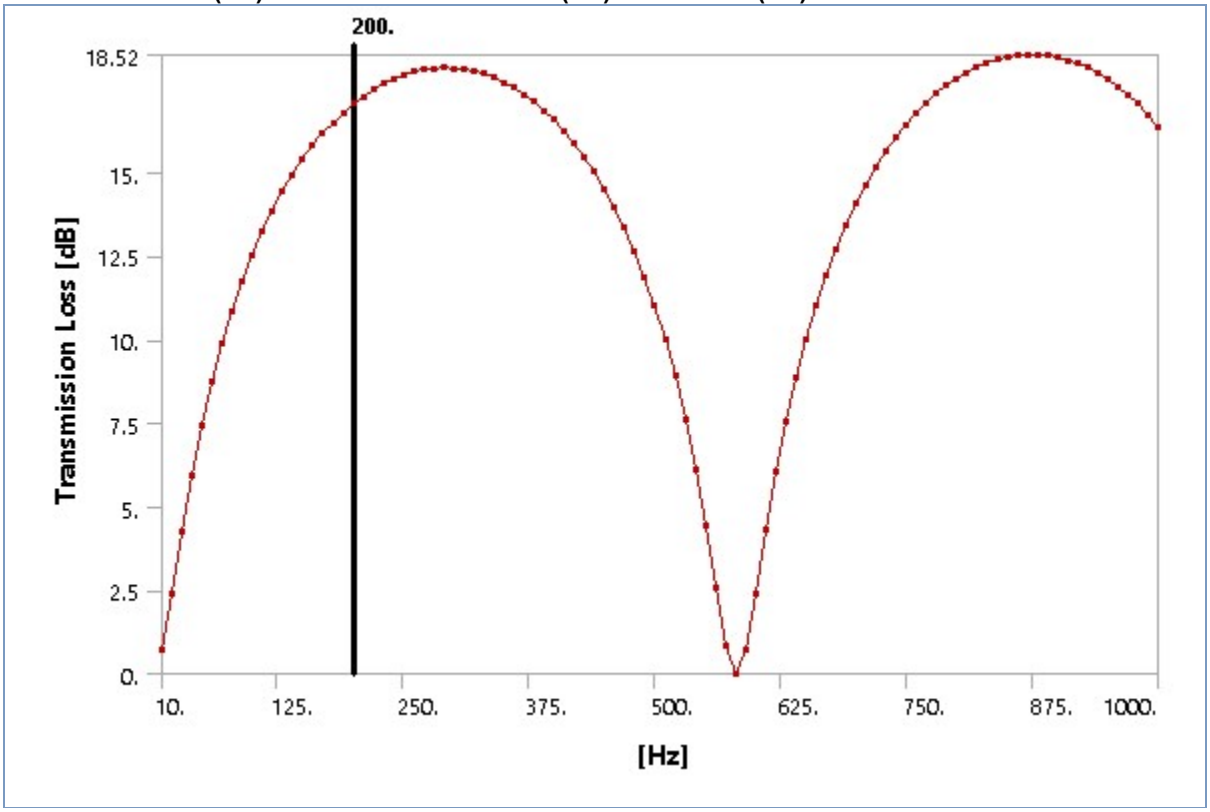


TABLE 18
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Transmission Loss

Frequency [Hz]	Transmission Loss [dB]
10.	0.74049
20.	2.4091
30.	4.2492
40.	5.9457
50.	7.4358
60.	8.7329
70.	9.8654
80.	10.861
90.	11.741

100.	12.524
110.	13.223
120.	13.851
130.	14.416
140.	14.924
150.	15.382
160.	15.794
170.	16.165
180.	16.497
190.	16.793
200.	17.056
210.	17.288
220.	17.489
230.	17.661
240.	17.806
250.	17.924
260.	18.016
270.	18.082
280.	18.123
290.	18.139
300.	18.13
310.	18.096
320.	18.037
330.	17.952
340.	17.841
350.	17.703
360.	17.538
370.	17.344
380.	17.119
390.	16.864
400.	16.575
410.	16.25
420.	15.886
430.	15.481
440.	15.031
450.	14.53
460.	13.974
470.	13.354
480.	12.662
490.	11.887
500.	11.014
510.	10.027
520.	8.9014
530.	7.6097
540.	6.1213
550.	4.4177
560.	2.5509
570.	0.82015
580.	0.
590.	0.71688
600.	2.4211
610.	4.3063
620.	

	6.0399
630.	7.5586
640.	8.8782
650.	10.029
660.	11.04
670.	11.934
680.	12.729
690.	13.441
700.	14.079
710.	14.654
720.	15.172
730.	15.639
740.	16.061
750.	16.441
760.	16.782
770.	17.087
780.	17.359
790.	17.599
800.	17.809
810.	17.99
820.	18.143
830.	18.27
840.	18.37
850.	18.445
860.	18.495
870.	18.52
880.	
890.	18.495
900.	18.444
910.	18.369
920.	18.267
930.	18.138
940.	17.982
950.	17.798
960.	17.583
970.	17.337
980.	17.058
990.	16.743
1000.	16.389

FIGURE 3
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Far-field SPL Mic

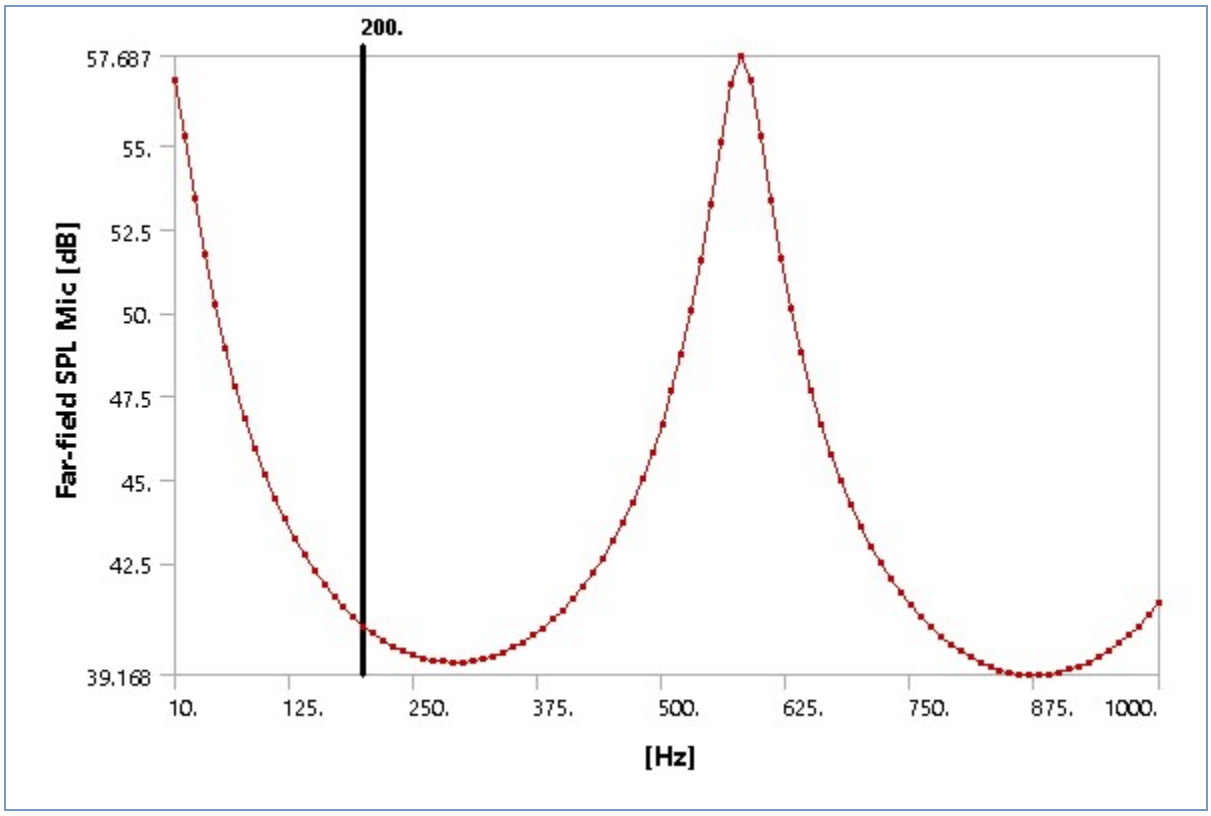


TABLE 19
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Far-field SPL Mic

Frequency [Hz]	Far-field SPL Mic [dB]
10.	56.947
20.	55.278
30.	53.438
40.	51.742
50.	50.252
60.	48.955
70.	47.822
80.	46.827
90.	45.947
100.	45.164
110.	44.464
120.	43.836
130.	43.272
140.	42.764
150.	42.306
160.	41.893
170.	41.523
180.	41.19
190.	40.894
200.	40.631
210.	40.4
220.	40.198
230.	40.026
240.	39.881
250.	39.763

260.	39.671
270.	39.605
280.	39.564
290.	39.548
300.	39.557
310.	39.591
320.	39.65
330.	39.735
340.	39.846
350.	39.984
360.	40.149
370.	40.343
380.	40.567
390.	40.823
400.	41.112
410.	41.437
420.	41.801
430.	42.205
440.	42.656
450.	43.157
460.	43.713
470.	44.333
480.	45.025
490.	45.8
500.	46.672
510.	47.66
520.	48.785
530.	50.077
540.	51.565
550.	53.269
560.	55.136
570.	56.866
580.	57.687
590.	56.97
600.	55.266
610.	53.38
620.	51.647
630.	50.128
640.	48.809
650.	47.658
660.	46.647
670.	45.753
680.	44.957
690.	44.246
700.	43.608
710.	43.033
720.	42.515
730.	42.048
740.	41.626
750.	41.247
760.	40.905
770.	40.6
780.	

	40.329
790.	40.089
800.	39.879
810.	39.698
820.	39.544
830.	39.418
840.	39.317
850.	39.243
860.	39.193
870.	39.168
880.	39.169
890.	39.194
900.	39.244
910.	39.32
920.	39.422
930.	39.551
940.	39.707
950.	39.892
960.	40.107
970.	40.353
980.	40.632
990.	40.948
1000.	41.301

FIGURE 4
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Far-field A-Weighted SPL Mic

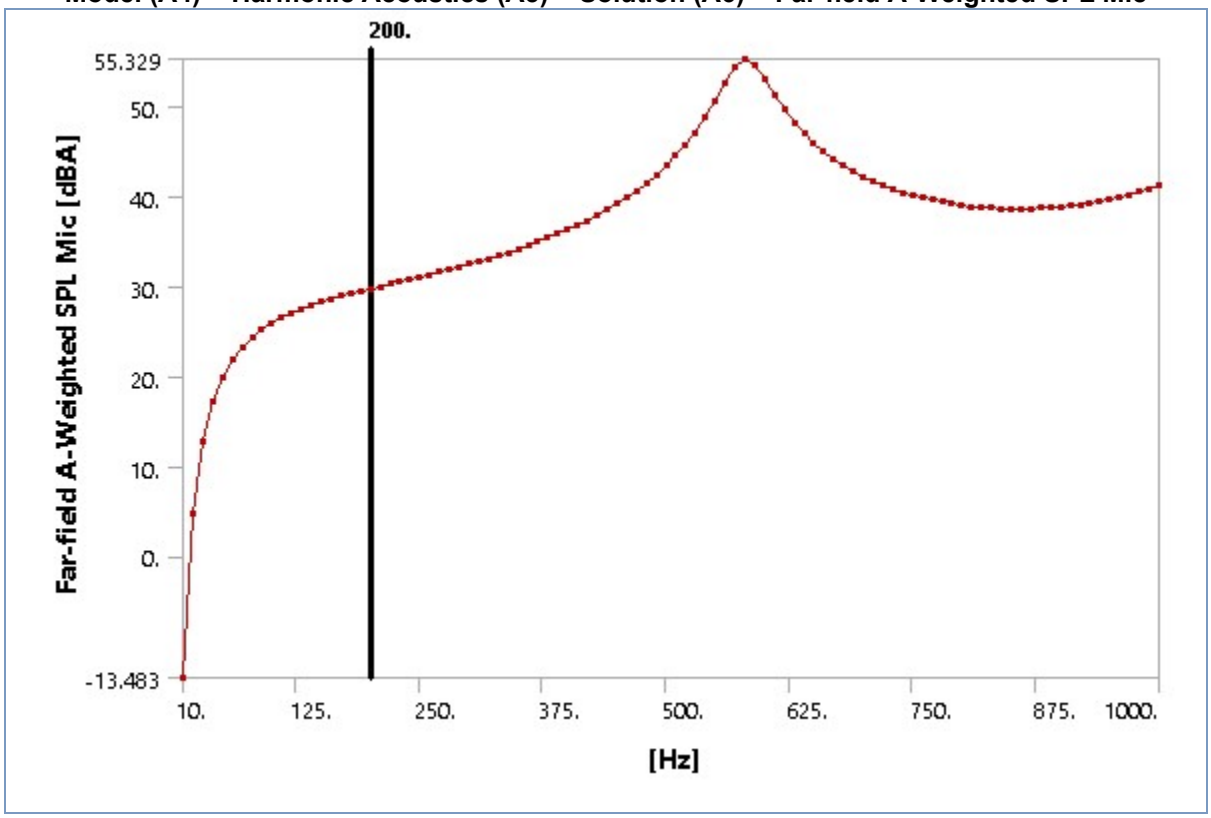


TABLE 20
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Far-field A-Weighted SPL Mic

Frequency [Hz]	Far-field A-Weighted SPL Mic [dBA]
10.	-13.483
20.	4.888
30.	12.836
40.	17.206
50.	19.98
60.	21.909
70.	23.334
80.	24.432
90.	25.306
100.	26.021
110.	26.619
120.	27.129
130.	27.574
140.	27.968
150.	28.324
160.	28.65
170.	28.954
180.	29.242
190.	29.518
200.	29.785
210.	30.047
220.	30.306
230.	30.565
240.	30.825
250.	31.089
260.	31.357
270.	31.631
280.	31.913
290.	32.203
300.	32.503
310.	32.814
320.	33.137
330.	33.474
340.	33.826
350.	34.194
360.	34.58
370.	34.986
380.	35.412
390.	35.863
400.	36.339
410.	36.843
420.	37.379
430.	37.95
440.	38.561
450.	39.216
460.	39.921
470.	40.685
480.	41.515
490.	42.423
500.	43.425
510.	44.537
520.	

	45.783
530.	47.191
540.	48.792
550.	50.604
560.	52.576
570.	54.409
580.	55.329
590.	54.707
600.	53.096
610.	51.3
620.	49.654
630.	48.22
640.	46.982
650.	45.911
660.	44.977
670.	44.158
680.	43.435
690.	42.795
700.	42.225
710.	41.717
720.	41.263
730.	40.859
740.	40.498
750.	40.178
760.	39.895
770.	39.646
780.	39.429
790.	39.242
800.	39.084
810.	38.954
820.	38.849
830.	38.77
840.	38.716
850.	38.687
860.	38.681
870.	38.699
880.	38.742
890.	38.807
900.	38.898
910.	39.012
920.	39.152
930.	39.317
940.	39.51
950.	39.729
960.	39.978
970.	40.258
980.	40.57
990.	40.917
1000.	41.301

Material Data

Air

TABLE 21
Air > Constants

Density	1.225e-009 kg mm^-3
Specific Heat	1.0064e+006 mJ kg^-1 K^-1
Thermal Conductivity	2.42e-005 W mm^-1 K^-1
Viscosity	1.7894e-011 MPa s

TABLE 22
Air > Molecular Weight

Molecular Weight kg kmol^-1
28.966

TABLE 23
Air > Lennard Jones Length

Lennard Jones Length mm
3711

TABLE 24
Air > Lennard Jones Energy

Lennard Jones Energy mJ
78600

TABLE 25
Air > Thermal Accom Coefficient

Thermal Accom Coefficient
0.9137

TABLE 26
Air > Velocity Accom Coefficient

Velocity Accom Coefficient mm s^-1
913.7

TABLE 27
Air > Formation Entropy

Formation Entropy mJ K^-1
1.9434e+008

TABLE 28
Air > Reference Temperature

Reference Temperature K
298.15

TABLE 29
Air > Critical Pressure

Critical Pressure mJ mm^-3
3.758

TABLE 30
Air > Critical Temperature

Critical Temperature K
132.3

TABLE 31
Air > Acentric Factor

Acentric Factor
3.3e-002

TABLE 32
Air > Critical Volume

Critical Volume mm^3
2.857e+006

TABLE 33
Air > Isotropic Relative Permeability

Relative Permeability
1

TABLE 34
Air > Absorption Coefficient

Absorption Coefficient mm^-1
0

TABLE 35
Air > Speed of Sound

Speed of Sound mm s^-1
3.4625e+005